Table 12. Pin assignment and description (continued)

Pin SO8N	Pin WLCSP12	Pin TSSOP20	Pin UFQFPN20	Pin name (function upon reset)	Pin type	I/O structure	Note	Alternate functions	Additional functions
4	F3	8	6	PA1	I/O	FT_a	-	SPI1_SCK/I2S1_CK, USART2_RTS_DE_CK, TIM17_CH1, USART1_RX, TIM1_CH2, I2C1_SMBA, EVENTOUT	ADC_IN1
4	F3	9	7	PA2	I/O	FT_a	-	SPI1_MOSI/I2S1_SD, USART2_TX, TIM16_CH1N, TIM3_ETR, TIM1_CH3	ADC_IN2, WKUP4,LSCO
-	F1	10	8	PA3	I/O	FT_a	-	USART2_RX, TIM1_CH1N, TIM1_CH4, EVENTOUT	ADC_IN3
-	F1	11	9	PA4	I/O	FT_a	-	SPI1_NSS/I2S1_WS, USART2_TX, TIM1_CH2N, TIM14_CH1, TIM17_CH1N, EVENTOUT	ADC_IN4, RTC_TS, RTC_OUT1, WKUP2
-	F1	12	10	PA5	I/O	FT_a	-	SPI1_SCK/I2S1_CK, USART2_RX, TIM1_CH3N, TIM1_CH1, EVENTOUT	ADC_IN5
-	F1	13	11	PA6	I/O	FT_a	-	SPI1_MISO/I2S1_MCK, TIM3_CH1, TIM1_BKIN, TIM16_CH1	ADC_IN6
-	E2	14	12	PA7	I/O	FT_a	-	SPI1_MOSI/I2S1_SD, TIM3_CH2, TIM1_CH1N, TIM14_CH1, TIM17_CH1	ADC_IN7
5	D1	15	13	PA8	I/O	FT_a	-	MCO, USART2_TX, TIM1_CH1, EVENTOUT, SPI1_NSS/I2S1_WS, TIM1_CH2N, TIM1_CH3N, TIM3_CH3, TIM3_CH4, TIM14_CH1, USART1_RX, MCO2	ADC_IN8
-	-	-	-	PA9	I/O	FT_f	(2)	MCO, USART1_TX, TIM1_CH2, TIM3_ETR, I2C1_SCL, EVENTOUT	-
-	-	-	-	PA10	I/O	FT_f	(2)	USART1_RX, TIM1_CH3, MCO2, TIM17_BKIN, I2C1_SDA, EVENTOUT	-
5	D1	16	14	PA11 [PA9]	I/O	FT_a	(2)	SPI1_MISO/I2S1_MCK, USART1_CTS, TIM1_CH4, TIM1_BKIN2	ADC_IN11
6	E2	17	15	PA12 [PA10]	I/O	FT_a	(2)	SPI1_MOSI/I2S1_SD, USART1_RTS_DE_CK, TIM1_ETR, I2S_CKIN	ADC_IN12
7	B1	18	16	PA13	I/O	FT_a	(3)	SWDIO, IR_OUT, TIM3_ETR, USART2_RX, EVENTOUT	ADC_IN13
8	C2	19	17	PA14-BOOT0	I/O	FT_a	(3)	SWCLK, USART2_TX, EVENTOUT, SPI1_NSS/I2S1_WS, USART2_RX, TIM1_CH1, MCO2, USART1_RTS_DE_CK	ADC_IN14, BOOT0
8	A2	20	18	PB6	I/O	FT_f	-	USART1_TX, TIM1_CH3, TIM16_CH1N, TIM3_CH3, USART1_RTS_DE_CK, USART1_CTS, I2C1_SCL, I2C1_SMBA, SPI1_MOSI/I2S1_SD, SPI1_MISO/I2S1_MCK, SPI1_SCK/I2S1_CK, TIM1_CH2, TIM3_CH1, TIM3_CH2, TIM16_BKIN, TIM17_BKIN	WKUP3
1	D3	1	19	PB7	I/O	FT_f	-	USART1_RX, TIM1_CH4, TIM17_CH1N, TIM3_CH4, I2C1_SDA, EVENTOUT, USART2_CTS, TIM16_CH1, TIM3_CH1, I2C1_SCL	RTC_REFIN

1. RST I/O structure when the PF2-NRST pin is configured as reset (input or input/output mode), FT I/O structure when the PF2-NRST pin is configured as GPIO
2. Pins PA9 and PA10 can be remapped in place of pins PA11 and PA12 (default mapping), using SYSCFG_CFGR1 register.
3. Upon reset, this pin is configured as SWD alternate function, and the internal pull-up on PA13 pin and the internal pull-down on PA14 pin are activated.